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INTRODUCTION

Key Concepts

Ant Colony, Artificial Neural Networks (ANN), Belief Propagation, Boundary Regions, Curve Fitting, Darwin, Evolutionary Search, Fuzzy Control, Fuzzy-evolutionary, Fuzzy Inference, Fuzzy Systems, Genetic Algorithms (GAs), Hybrid Systems, Imprecision, Inexactness, Information Systems, Law of Excluded middle, Learning, Measurement, Natural Selection, Neuro-fuzzy, Neuro-genetic, Optimization, Pattern Association, Pattern Classification, Pattern Clustering, Perception, Probabilistic Reasoning, Rough Sets, Simulated Annealing (SA), Soft Computing, State Space Search, Supervised Learning, Survival of the Fittest, Swarm Optimization, Training Pairs, Unsupervised Learning, Vagueness.

Chapter Outline

1.1 What is Soft Computing?
1.2 Fuzzy Systems
Rough Sets
Artificial Neural Networks
Evolutionary Search Strategies

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What is soft computing? How does it differ from traditional ‘hard’ computing? What are its main components and how do they relate to each other? What are the basic traits of a soft computing technique? These are relevant questions that any inquisitive mind will ask while approaching a text on the subject. While clear answers to these questions can only emerge out of careful study and practice of the subject, some indicators need to be available to the sincere learner at the very outset. This chapter is intended to provide some ideas about the spirit of soft computing as a computational process.

1.1 WHAT IS SOFT COMPUTING?

Well, the obvious answer to this question is, ‘*soft computing is computing which is not hard*’. But this candid answer might not satisfy some people who are so inherently skeptic as to ask further, ‘*then what is hard computing?*’ The rest of this chapter is dedicated to these skeptics.

The phrase ‘soft computing’ was coined by Lotfi A. Zadeh, a pioneer in this field, in the early 1990s. The question ‘What is soft computing?’ can be answered from various perspectives, e.g., its methodological traits, problem solving abilities, constraints, and so on. In fact a complete understanding of the term ‘soft computing’ is possible only when all these perspectives are taken into consideration. In this introductory text, we will certainly try our best to build this holistic view of soft computing, but presently we start by focusing our attention on what is readily available to us, the phrase ‘soft computing’.

The term ‘soft computing’ consists of two words, ‘soft’ and ‘computing’. Assuming that we have a fairly good idea of what ‘computing’ is, let us focus our attention on the remaining keyword, i.e., ‘soft’ which is opposite to ‘hard’. Table 1.1 presents certain characteristics that we tend to associate with these two contrasting words.

Table 1.1. *Hard vs. soft*

Hard	Soft
Rigid	Flexible
Fixed	Movable/Adjustable
Systematic	Random
Well-defined	Vague
Exact	Inexact/Approximate
Precise	Imprecise
Measurable	Perceivable
Solid	Porous
Deterministic	Non-deterministic
...	...

Taking a cue from the words closely associated with ‘hard’ and ‘soft’, we may expect that soft computing somehow relates to flexibility, imprecision, inexactness, vagueness, randomness, non-determinism and so on either as computational process, or the computational problems they try to solve. Indeed, these traits distinguish soft computing from hard computing.

In real life we keep on confronting and handling situations characterized by soft qualities. Let us consider a few.

- (a) *Parking a car on a narrow parking space*. You want to park a car within a narrow parking space. The available space is just enough to keep the car. You don’t need to measure the exact length or breadth of the space. Nor you need to know the exact coordinates of the car’s final position. However, you successfully assess the situation and maneuver the car in a way such that the car is finally parked properly.
- (b) *Recognition of handwritten characters*. There are infinite variations in the shape of an alphanumeric character written by people. None of them exactly match the printed character. In printed form too, the same character have different shapes in different fonts. Add to this the variation due to size, writing material, colour, the surface on which the characters are written etc. In spite of all these deviations from the ‘ideal’ shape, we have no difficulty in recognizing a handwritten character. It seems that our brain do not process the image of such an entity pixel by pixel but as whole. This is again in sharp contrast with the conventional hard computing paradigm.

- (c) *Collection of food by ants.* When the ants look for food they start by traveling in random directions. Many ants go out of the ant-hill simultaneously and individually search for food in various directions. However, when the source of food is discovered by one or a few ants, they return to their ant-hill and convey this message to other ants. Gradually, the movements of the entire population of the ants (i.e. those who are engaged in searching and collecting food) converge to a line joining the source of food and the ant-hill. And, this line is usually along the shortest distance between the source and the destination.

Situation (a), that of parking a car on a narrow parking space, is an instance of a problem whose description is imprecise in the sense that the exact measurement of the dimensions of the parking space is not available. In fact, such exact values are neither necessary, nor desirable because it is not a precision job at all. Also unavailable are the dimensions of the car (we do not bother about those figures while parking), and the coordinates of the wheels when the car is finally parked. This is a typical situation where exactitude, or precision, in the description of the problem as well as the solution is neither available, nor necessary.

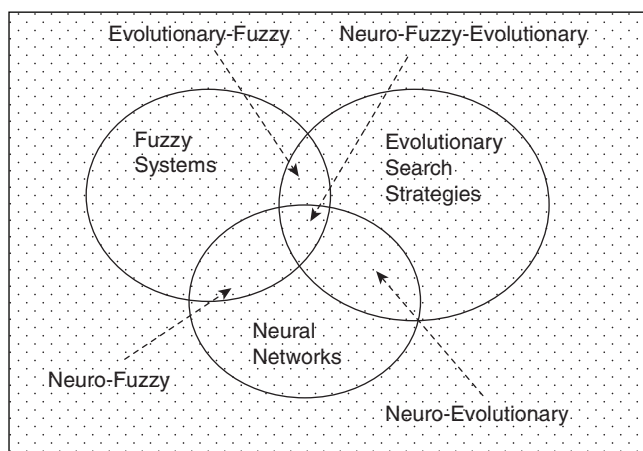
Situation (b), that of recognition of handwritten characters, exemplify the distinctive nature of human perception as opposed to traditional computational process based on instruction fetch and execute cycle. It seems that the human brain perceives a pattern as a whole, not pixel by pixel. Moreover, small deviations, or incompleteness of description, or similar aberrations from the ideal pattern do not prevent us from recognizing the pattern correctly. This remarkable capacity of human brain is the result of the structure of the brain that allows immense parallelism. There is another important phenomenon called learning. Conventional computation does not model learning. Rather, it is based on the idea of an algorithm as an embodiment of procedural knowledge already learnt by the programmer.

The third situation, the behaviour of ants while searching for food, is an instance of nature's wisdom to achieve betterment over time. This is similar to an optimization process. A more elaborate optimization process undertaken by nature is the evolution of higher order species from lower ones over millions and millions of years by means of natural selection. Randomness is a necessary ingredient of these processes. However, such randomness is not unconstrained. Randomness is necessary to explore possibilities, but at the same time, it must be supported by direction. This is ensured by various mechanisms. For example, in biological evolution the upthrust is provided by natural selection guided by Darwinian principle of survival of the fittest. In ant colonies, this is provided by accumulation of a chemical called pheromone deposited by the ants along the frequently visited paths. Moreover, the target of optimization is also 'softened' in these processes. This is because, unlike traditional optimization, here we do not insist on 'the' optimal solution because we may have to wait too long to receive the best solution. Rather, a near-optimal solution available at a convenient time is accepted. The fact is that certain problems are computationally so complex that finding the best solution would take ages by even the fastest computer. For most practical purposes, a quickly available near-optimal solution at the expense of a slight, probable, compromise in quality is acceptable.

What is soft computing then? It is not a single computational technique. Soft computing is a family of techniques with capacity to solve a class of problems for which other conventional techniques are found to be inadequate. The principal components of soft computing, as on today, includes fuzzy systems (fuzzy set theory, fuzzy logic, fuzzy inference systems etc.), rough set theory, artificial neural networks, probabilistic reasoning, and evolutionary search strategies (including genetic algorithms, simulated annealing, ant colony optimization, swarm optimization etc.). Table 1.2 provides a summary of the domains of these components of soft computing.

Table 1.2. *Soft Computing Techniques*

#	Technique	Application domain
1	Fuzzy systems	Vagueness / imprecision / inexactness / Approximate reasoning
2	Rough sets	Vagueness / inexactness in information systems
3	Artificial neural networks	Learning and curve fitting / Pattern classification, association, clustering
4	Probabilistic reasoning	Uncertainty and belief propagation
5	Evolutionary searches	Complex optimization

**Fig. 1.1.** *Synergy among the principal components of soft computing*

It should be noted that while each of these techniques can be applied in isolation to solve problems of related domains, they can work together synergistically. The fact is, soft computing is not just a collection of several techniques, but is a family of highly interacting and complementary techniques.

For instance, artificial neural networks generally lack certain characteristics which are present in fuzzy logic. On the other hand, fuzzy systems cannot learn, adapt, or support parallelism though these are clearly present in neural nets. This observation prompted researchers to develop neuro-fuzzy systems that are highly successful hybrid systems. The complementary role of fuzzy logic and neuro-computing helps a neuro-fuzzy system overcome the limitations of both constituents. Actually, hybridization is a central theme of soft computing. Various hybrid soft computing systems, e.g., neuro-fuzzy systems, fuzzy neural networks, genetic fuzzy systems, fuzzy-evolutionary algorithms, genetic-neural networks etc. have been developed in past years and are being developed. Fig. 1.1 gives a graphical view of hybridization in soft computing.

What are the essential properties that bring all these diverse methodologies together under the common umbrella named 'soft computing'? We can safely say that a computation that deliberately incorporates imprecision on one or more levels of computation resulting either in a change, in fact decrease, in the 'granularity' of the problem, or relaxing the goal of optimization at some stage, is a kind of soft computing. So the effect of including imprecision is a relaxation either in the level of description of the

problem, or the level of achievement of the goal. However, it should be noted that the imprecision is not a target. It is a means to achieve practical solution to a given complex problem.

Soft computing can also be considered as a tool to tackle imprecision and uncertainty. As stated by Zadeh, the guiding principle of soft computing is to exploit the tolerance for imprecision, uncertainty, partial truth, and approximation to achieve tractability, robustness, low cost etc. Real life problems are full of uncomfortable characteristics due to partial, vague, noisy, and incomplete information. Under such circumstances, hard computing techniques are not appropriate. In this context, it is worthwhile to ponder over the difference between measurement and perception. While measurements are exact, perceptions are vague. We, the humans, have a wonderful capacity to perform numerous tasks, physical as well as mental, without any measurement or calculations. While science has been motivated to progress from perception to measurement, soft computing perhaps will enable us to return to perception. Hence soft computing may lead in future to computing with words and perceptions.

In the rest of this chapter, we will try to appreciate the spirit of soft computing by a quick review of the basic principles of its components.

1.2 FUZZY SYSTEMS

Fuzzy set theory is an extension, or generalization, of the classical, occasionally referred to as crisp, set theory. Our everyday conversation is full of vague and imprecise concepts, e.g., ‘Uncle Sam is tall’ or ‘It is very cold today’. It is difficult if not impossible to translate such statements into more precise language because such an effort results in losing some of their semantic values. If, instead of saying ‘It is very cold today’ someone says ‘Today’s temperature is 5°C’, or, instead of saying ‘Uncle Sam is tall’ we say ‘Uncle Sam’s height is 5 ft 10 inch’, aren’t we losing some of the meanings of the statements mentioned above? It seems that in real life, vagueness is not only unavoidable but also desirable to some extent.

As stated earlier, science tries to replace perception with measurement. However, more and more we are facing situations in science and technology where there is a need to retain perception into the system. For instance, consider the case of designing an expert system to embody the diagnostic power of a physician. In his effort to codify the physician’s decision making process, the designer discovers that the physician’s approach to diagnosis or medication is largely intuitive though supported by various test results and measurements. Accepting vagueness as a valid element of model of reality, instead of trying to mould reality into crisp measurements, is the natural way to implement such applications.

Fuzzy set theory, which effectively models vagueness in its mathematical formulation, is naturally extended to logic. Logic is the study of the structures and principles of reasoning and sound argument. Classical logic is based on the famous ‘law of excluded middle’. This law states that every statement must be either true or false. There was strong opposition to this view even in ancient times. Greek philosopher Heraclitus opined that statements could be simultaneously true and false. The central theme of fuzzy logic was upheld by Plato who indicated that there was a third region beyond true or false. In recent past, the systematic alternative to bi-valued logic of Aristotle was proposed by Lukasiewicz in early 1900s. Lukasiewicz proposed a 3-valued (true, false, and ‘possible’) logic, followed by 4-valued and later 5-valued logics. The modern, infinite-valued fuzzy logic was proposed by LA Zadeh in 1965. In real life we face situations where there is no sharp distinction between truth and falsehood. Rather, there are infinite shades of truths between absolute truth and absolute falsehood. Fuzzy logic accepts this state of affair and builds a system of reasoning on the basis of infinite shades of truth. Fuzzy logic is one of the most successful theories in terms of practical applications. An important class of applications is based on the idea of fuzzy inference system. This is a kind of input-output mapping based on

fuzzy logic. These systems have been applied in machine control and are popularly known as fuzzy control systems. The advantage of fuzzy inference systems is that here the solution to the problem can be cast in terms of familiar human operators. Hence, the human experience can be used in the design of the controller. Engineers developed numerous fuzzy controllers for industrial applications and consumer products.

1.3 ROUGH SETS

Fuzzy sets and fuzzy logic have modeled vagueness in terms of partial membership and partial truth respectively. Another model of vagueness has gained importance in recent past. The concept of rough sets, proposed by Z Pawlak, considers vagueness from a different perspective. Here vagueness is expressed, instead of set membership, in terms of *boundary regions* of a set of *objects*. In large reservoir of multidimensional data, occasionally it is not possible to decide with certainty whether a given object belongs to a set or not. Such objects are said to form a *boundary regions* for the set. If the boundary region is empty, then the set is crisp, otherwise it is *rough*. A non-empty boundary region exists due to insufficient knowledge to define the set with certainty. There are many interesting applications of the rough set theory including knowledge acquisition, decision analysis, knowledge discovery from databases, expert systems, inductive reasoning etc.

1.4 ARTIFICIAL NEURAL NETWORKS

A computer program embodies a readymade procedural knowledge that the programmer has acquired and then translated with help of a programming language. Superiority of the brain over computer is largely ascribed to brain's capacity to learn from experience. The slow process of learning enables man to perform certain tasks, e.g., recognition, classification, association, clustering etc. in a highly efficient manner.

Artificial neural networks (ANNs) are inspired by the structure and functionality of the brain. There are nearly 100 billion neurons in a normal human brain. Each neuron is locally connected to its neighbouring neurons. The neurons have elementary capacities like summing up the incoming signals and then passing it on to the neighbours conditionally. Human consciousness is the outcome of the collective activities of these 100 billion neurons. In a computer information is stored as localized bits. An ANN preserves information as weights of interconnections among its processing units. Thus, as in the brain, information in ANNs too resides in a distributed manner, resulting in greater fault tolerance. Moreover, multiple data may be superimposed on the same ANN for storage purpose. Like the human brain, ANNs also perform computation in terms of patterns rather than data.

Pattern classification is the task of deciding whether the input pattern, usually a vector, belongs to a certain class or category. In real life, we encounter pattern classification tasks quite often. For instance, we may need to classify a cell, on the basis of its image, as cancer-affected or otherwise.

Another common human experience is pattern association. It takes place when we relate a given pattern to one already stored in memory. We do associate patterns in our daily life almost without any conscious effort. Examples are, recognition of a known face from an image (either distorted, or undistorted), visualizing a flower from its fragrance, remembering a distant past on hearing a particular tune etc. In computing, retrieval of a stored pattern corresponding to an input pattern is known as pattern association. Associative memory neural nets are those which store a set of pattern associations. There are two kinds of associative memory neural nets. The associations may be auto (the input and the stored patterns are identical), or hetero (the input and the stored patterns are different).

Neural nets are trained by providing sample classification data over and over again and making adjustments in their weight vectors so that they become 'experienced' enough to classify unknown patterns successfully. Learning is either supervised or unsupervised. When learning takes place in presence of a teacher the learner has the opportunity to get corrected by the teacher when he commits a mistake. This is supervised learning. A self-learner, in contrast, does not have a teacher and therefore he has to identify an error as well as get it corrected all by himself. Similarly, learning by ANN is either supervised or unsupervised, depending on the availability of training data. By training data we mean a set of pairs of input–output vectors. In presence of such data, the ANN can measure the deviation from desired output values when the net is presented with an input pattern. Supervised learning by an ANN takes place in this way. However, in the absence of such training pairs, the ANN has to adjust itself on its own. Usually some kind of competition facilitates unsupervised learning. There are various ANNs, e.g., Kohonen's self organizing map (SOM), learning vector quantization (LVQ), etc. that act on the basis of unsupervised learning. This text contains discussions on the fundamental ANNs including Hebb nets, Perceptrons, ADALINE, MADALINE etc. as pattern classifiers, Hopfield nets and bidirectional associative memory (BAM) as associative networks, Kohonen's self organizing map (SOM), learning vector quantization (LVQ), adaptive resonance theory (ART) as competitive networks, backpropagation networks, etc.

1.5 EVOLUTIONARY SEARCH STRATEGIES

Quite often, intelligent computing takes the form of a state space search. A state space is a graph where the nodes represent the 'states' relating to a computational problem and the directed edges represent possible moves from one problem state to another. Starting with the initial state, the requirement is to reach a 'goal' state by traversing a suitable path through the graph. The state spaces are often huge in size and finding a solution, or a path to the solution, may prove to be highly computation intensive process. In this text we discuss the exhaustive search techniques, e.g., breadth-first search, depth-first search, depth-first iterative deepening etc., as well as various heuristic search strategies. The principles underlying best-first search, hill climbing, A / A* algorithm, AO* algorithm etc. are presented with illustrative examples.

Complex optimization problems require advanced search techniques to obtain workable solutions within reasonable time frame. Classical optimization techniques can be used only on continuous and differentiable functions. However, often such well behaved functions are not available for certain optimization problems. Moreover, classical search techniques have a tendency to settle down at local optima instead of the global best. There are computational problems which require tremendous computational efforts to find the best solution. Intelligent search strategies like hill climbing may be employed to obtain reasonably good solutions to such problems. However, hill climbing suffers from the serious problem of settling to sub-optimal solutions remaining in the search space as local optimal points. Genetic Algorithms (GAs) and Simulated Annealing (SA) are two search strategies that are inspired by natural evolutionary processes and have the capacity to overcome the problem posed by the existence of local optima in large search spaces. GAs are inspired by the process of natural evolution. The mechanism applied by nature in evolution is natural selection based on the Darwinian principle of survival of the fittest. It is essentially a maximization process. Simulated Annealing (SA) mimics the process of physical annealing. In physical annealing a metal is initially heated to a molten state and then gradually cooled to get a uniform crystal structure. This uniform crystal structure corresponds to a minimal energy level. Hence annealing is a minimization process. The GAs and SAs are extensively applied to solve optimization problems of highly complex nature.

**CHAPTER SUMMARY**

The forgoing introductory discussion on the nature and constituents of soft computing can be summarized in the following way.

- The term ‘soft computing’ was coined by LA Zadeh in early 1990s and can be interpreted from various perspectives, e.g., its methodological traits, problem-solving abilities, constraints, and so on. A complete understanding of the term ‘soft computing’ is possible only when all these perspectives are taken into consideration.
- Soft computing is a family of techniques with capacity to solve a class of problems for which other conventional techniques are found to be inadequate. The principal components of soft computing include fuzzy systems, rough set theory, artificial neural networks (ANNs), probabilistic reasoning, and evolutionary search strategies including genetic algorithms (GAs), simulated annealing (SA), ant colony optimization, swarm optimization etc.
- Fuzzy systems are systems built on fuzzy set theory and fuzzy logic. These systems try to model vagueness, or inexactness, which is a necessary ingredient of everyday interactions and activities. Fuzzy set theory accommodates vagueness by allowing set membership values to lie anywhere between 0 and 1, both inclusive. Fuzzy logic violates the Aristotelian law of excluded middle and allows a statement to be true to any extent between absolute falsehood and absolute truth.
- In rough set theory, vagueness is expressed, instead of set membership, in terms of boundary regions of a set of objects. If the boundary region is empty, then the set is crisp, otherwise it is rough.
- Artificial neural nets are networks of processing elements that follow a computational paradigm akin to that of the human brain. These are efficient structures to classify, associate and cluster patterns. Like human brain, the artificial neural nets need to be trained to carry out the designated task. Learning by ANN could be either supervised, or unsupervised. Supervised learning is assisted by training data and unsupervised learning takes place in absence of any training data.
- Complex optimization problems require advanced search techniques to obtain workable solutions within reasonable time frame. Genetic Algorithms (GAs) and Simulated Annealing (SA) are two search strategies that are inspired by natural evolutionary processes and have the capacity to overcome the problem posed by the existence of local optima in large search spaces. While GA is a maximization process, SA is a minimization process.

**TEST YOUR KNOWLEDGE**

- 1.1 Which of the following traits is expected in a soft computing technique?
 - a) Precision measurement
 - b) Exactitude
 - c) Absolute truth/falsehood
 - d) None of the above
- 1.2 Which of the following traits is not expected in a soft computing technique?
 - a) Randomness
 - b) Softening of goal
 - c) Vagueness
 - d) None of the above
- 1.3 Fuzzy logic is a soft computing technique to deal with
 - a) Vagueness
 - b) Learning
 - c) Optimization
 - d) None of the above

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